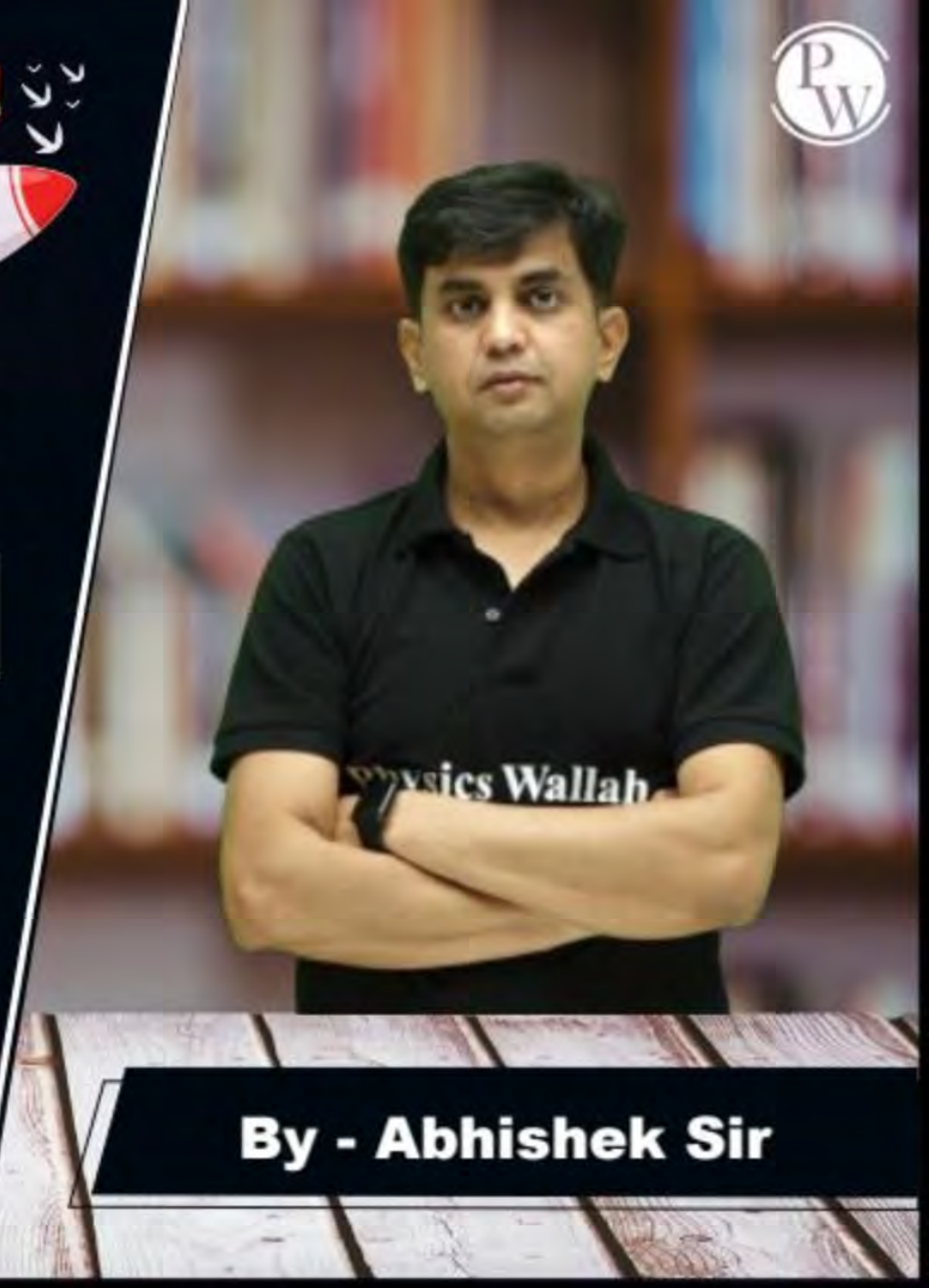
CS & IT ENGINEERING

Computer Network - 2

Network Layer

Lecture No. - 03



By - Abhishek Sir

Recap of Previous Lecture

Topic IPv4 Options

Topic ICMP

Topics to be Covered

Topic Supernetting

Topic NAT

[MCQ] [GATE-IT-2005] [1 Mark]

Tracert

#Q. Trace-route reports a possible route that is taken by packets moving from some host A to some other host B, which of the following options represents the technique used by traceroute to identify these hosts?

A. ~~By progressively querying routers about the next router on the path to B using ICMP packets, starting with the first router.~~

B. ~~By requiring each router to append the address to the ICMP packet as it is forwarded to B. The list of all routers en-route to B is returned by B in an ICMP reply packet.~~ ⇒ Record-Route

C. By ensuring that an ICMP reply packet is returned to A by each router en-route to B, in the ascending order of their hop distance from A

D. ~~By locally computing the shortest path from A to B~~

$\boxed{\text{Ans: C}}$

Topic : Supernetting

CS-22

→ Prefix Aggregation / CIDR Aggregation / Route Aggregation

→ Combining (logically) smaller networks into single large network

CS-23

CIDR Aggregation

Topic : Supernetting

suppose 8-bit IP Address;

Example 1 : Suppose network id field size are 6 bits and host id field size are 2 bits. Consider following Network Addresses of networks, what should be the supernet address?

Net. Add.1	:	1 0 1 0 0 0 0 0 / 6
Net. Add.2	:	1 0 1 0 0 1 0 0 / 6
Net. Add.3	:	1 0 1 0 1 0 0 0 / 6
Net. Add.4	:	1 0 1 0 1 1 0 0 / 6

Supernet Address1: 1 0 1 0 0 0 0 0 / 5

Supernet Address2: 1 0 1 0 1 0 0 0 / 5

Supernet Address : 1 0 1 0 0 0 0 0 / 4

Topic : Supernetting

Supernet Address : 10100000/4

Example 1 :

Supernet Address₁ : 10100000/5

Supernet Address₂ : 10101000/5

Net. Add.1 : 1 0 1 0 0 0 0 0 / 6
1 0 1 0 0 0 0 1
1 0 1 0 0 0 1 0
1 0 1 0 0 0 1 1

Net. Add.2 : 1 0 1 0 0 1 0 0 / 6
1 0 1 0 0 1 0 1
1 0 1 0 0 1 1 0
1 0 1 0 0 1 1 1

Net. Add.3 : 1 0 1 0 1 0 0 0 / 6
1 0 1 0 1 0 0 1
1 0 1 0 1 0 1 0
1 0 1 0 1 0 1 1

Net. Add.4 : 1 0 1 0 1 1 0 0 / 6
1 0 1 0 1 1 0 1
1 0 1 0 1 1 1 0
1 0 1 0 1 1 1 1

Topic : Supernetting

supernet Address : $\boxed{10100000/4}$

Example 1 :

Net. Add.$_1$: 10100000 / 6
1010 0001
1010 0010
1010 0011

Net. Add.$_2$: 10100100 / 6
1010 0101
1010 0110
1010 0111

Net. Add.$_3$: 10101000 / 6
1010 1001
1010 1010
1010 1011

Net. Add.$_4$: 10101100 / 6
1010 1101
1010 1110
1010 1111

Topic : Supernetting

Example 2 :- Consider following Network Addresses of networks. What should be supernet address ?

150 . 125 . 160 . 0 / 23

150 . 125 . 162 . 0 / 23

150 . 125 . 164 . 0 / 23

150 . 125 . 166 . 0 / 23

Supernet Address : 150.125.160.0/21

$Net\ Add_1 = 150.125.160.0/23$

$1^{st}\ IP \rightarrow 150.125.160.0/23$

$last\ IP \rightarrow 150.125.161.255/23$

$\Big\}\ 2^9$ Address

$Net\ Add_2 = 150.125.162.0/23$

$1^{st}\ IP \rightarrow 150.125.162.0/23$

$last\ IP \rightarrow 150.125.163.255/23$

$\Big\}\ 2^9$ Address

$\rightarrow 2^{10}$ Address

Supernet Address$_1$: $150.125.10100000.00000000/22$

: $150.125.160.0/22 \longrightarrow 2^{10}$ Address

Net = 150.125.164.0/23

1st IP → 150.125.164.0/23

last IP → 150.125.165.255/23

$\}$ 2^9 Address

Net = 150.125.166.0/23

1st IP → 150.125.166.0/23

last IP → 150.125.167.255/23

$\}$ 2^9 Address

→ 2^{10} Address

Supernet Address$_2$: 150.125.10100100.00000000/22

150.125.164.0/22 ⟶ 2^{10} Address

Topic : Supernetting

Example 2 :- 23 bit prefix   9 bit Host ID

150 . 125 . 1 0 1 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 23 — 2^9

150 . 125 . 1 0 1 0 0 0 1 0 . 0 0 0 0 0 0 0 0 / 23 — 2^9

150 . 125 . 1 0 1 0 0 1 0 0 . 0 0 0 0 0 0 0 0 / 23 — 2^9

150 . 125 . 1 0 1 0 0 1 1 0 . 0 0 0 0 0 0 0 0 / 23 — 2^9

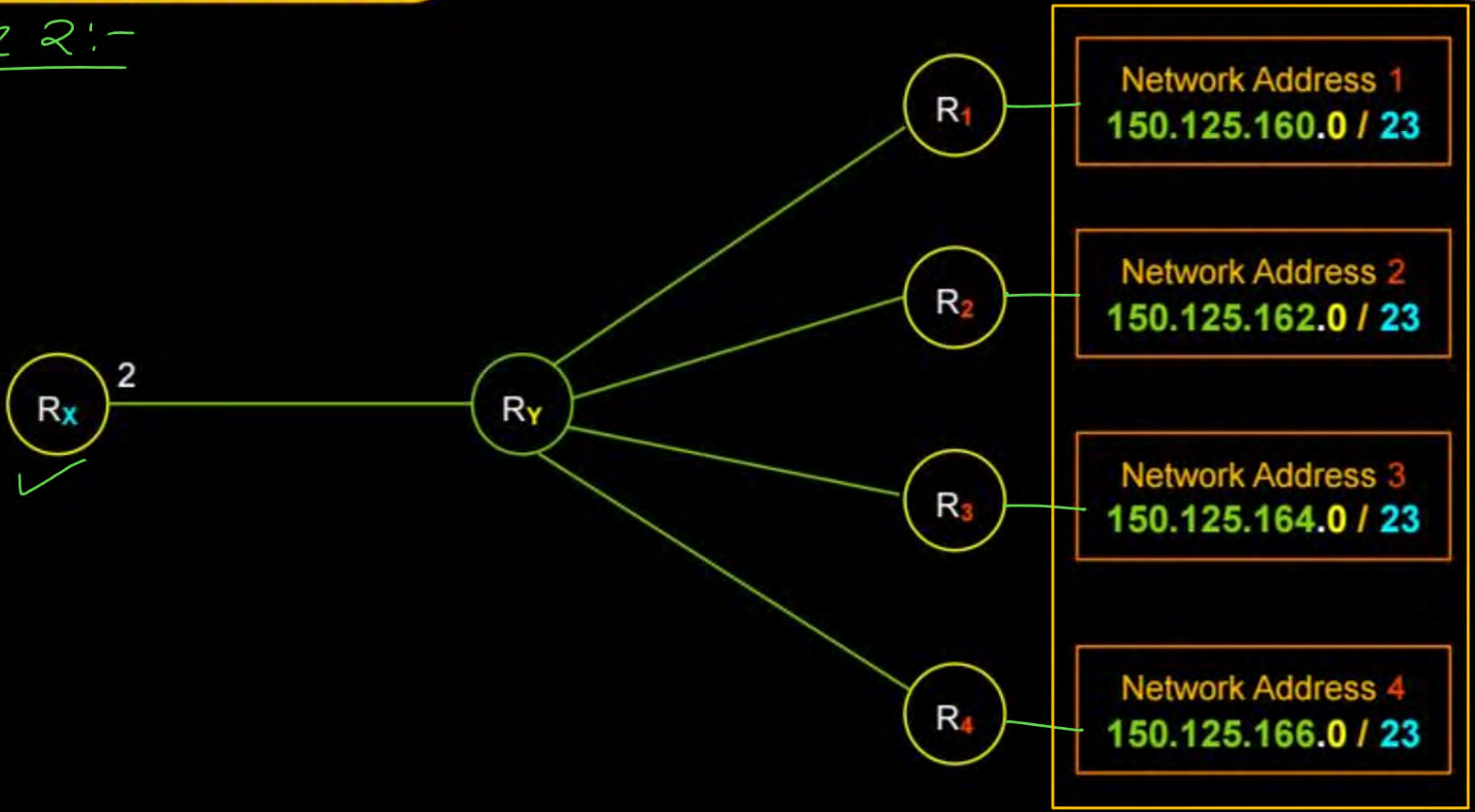
Supernet Address : 150 . 125 . 160 . 0 / 21

$\underbrace{150 . 125 . 1 0 1 0 0}_{\text{21 bit prefix}}\ \underbrace{0 0 0 . 0 0 0 0 0 0 0 0}_{\text{11 bit host ID}}$ /21 $\longrightarrow 2^{11} = (2^2 * 2^9)$ Address

Topic : Supernetting

2-bit supernetting

Example 2:-



Topic : Supernetting

Router (R_X) Forwarding (Routing) table

Network Address	Interface ID	Next Hop
150.125.160.0 / 23	2	R_Y
150.125.162.0 / 23	2	R_Y
150.125.164.0 / 23	2	R_Y
150.125.166.0 / 23	2	R_Y

Router (R_X) Forwarding (Routing) table

After Route (Prefix / CIDR) Aggregation

Network Address	Interface ID	Next Hop
150.125.160.0 / 21	2	R_Y

Topic : Supernetting

→ Reduces the number of entries in routing table (if possible) due to supernetting

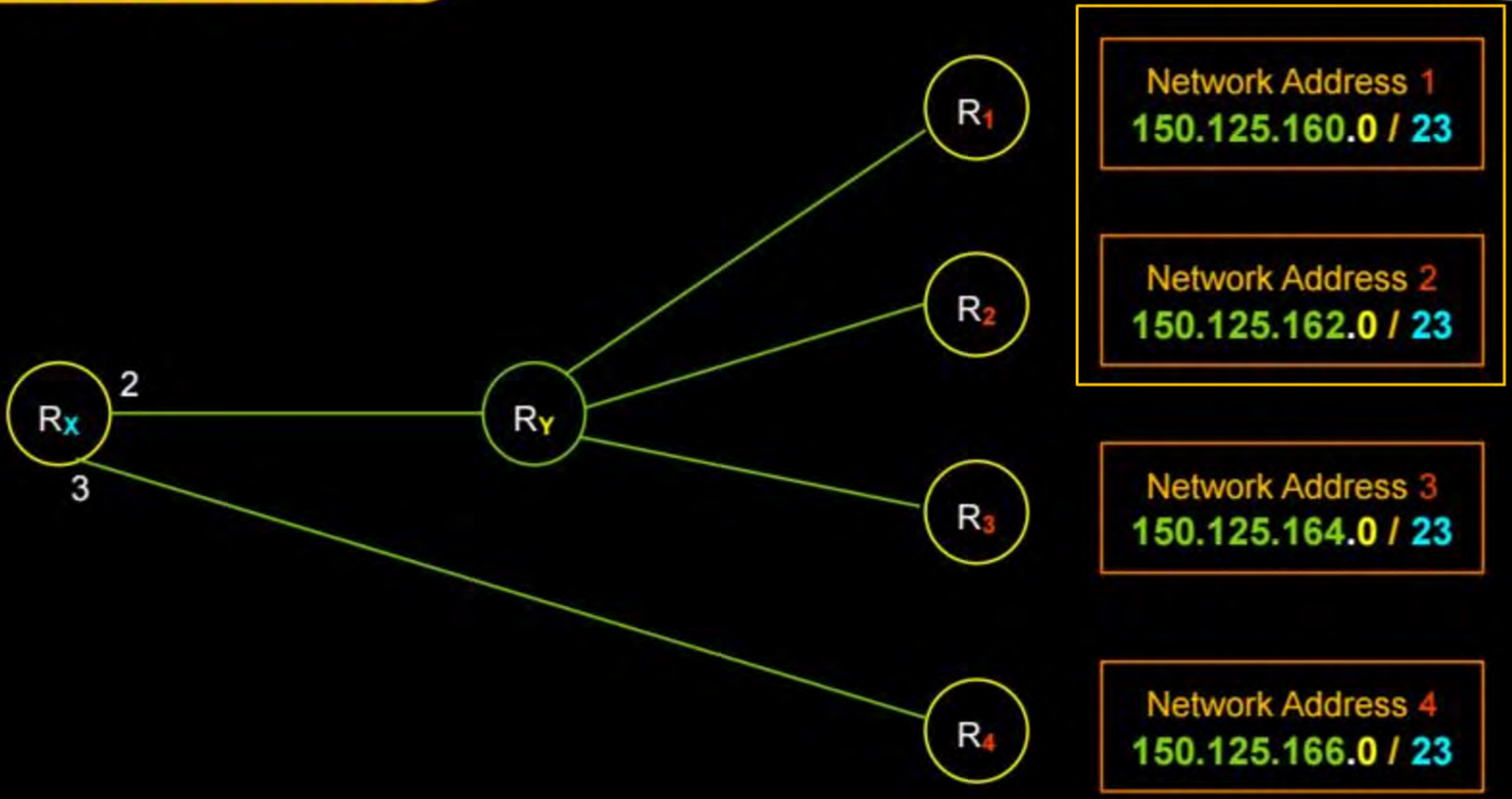
→ Allow more efficient routing ✓

→ All the Networks should be contiguous
[Block of addresses having contiguous prefixes]

⇒ Supernetting is a prog. run inside every router to optimise routing table

Topic : Supernetting

1-bit supernetting



Router (R_X) Forwarding (Routing) table

Network Address	Interface ID	Next Hop
150.125.160.0 / 23	2	R_Y
150.125.162.0 / 23	2	R_Y
150.125.164.0 / 23	2	R_Y
150.125.166.0 / 23	3	R_4

Router (R_X) Forwarding (Routing) table
After Route (Prefix / CIDR) Aggregation

Network Address	Interface ID	Next Hop
150.125.160.0 / 22	2	R_Y
150.125.164.0 / 23	2	R_Y
150.125.166.0 / 23	3	R_4

Topic : Supernetting

H.W.

Example 3 :- Consider following Network Addresses of networks.

210 . 192 . 0 . 0 / 13

210 . 200 . 0 . 0 / 13

210 . 208 . 0 . 0 / 13

210 . 216 . 0 . 0 / 13

Supernet Address : ?

Topic : Supernetting

H.W.

Example 4 :- Consider following Network Addresses of networks.

10 . 90 . 0 . 0 / 16

10 . 90 . 64 . 0 / 18

10 . 90 . 192 . 0 / 18

Supernet Address : ?

[MSQ] IIT-KGP, H.W. [GATE-2022][2 Mark]

#Q. Consider routing table of an organization's router shown below:

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

(A) 12.20.164.0/20

(B) 12.20.164.0/22

(C) 12.20.164.0/21

(D) 12.20.168.0/22

Topic : IPv4 Address

2^{32} Address

→ Solution for IPv4 address (32-bits) range problem.

1. Network Address Translation (NAT) ⟹ ✓ (CS-24) IISC
   [Short-term solution]

2. IPv6 address (128 bits) ⟹ 2^{128} Address
   [Permanent solution]

2 mins Summary

Topic Supernetting

Topic NAT

DHCP

Traffic shaping

THANK - YOU

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